Shan-Cheng Chang 張善程

AI Application R&D Engineer · ships LLM application systems to production

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SUMMARY

AI application engineer (M.S., NTU) with nearly two years taking LLM systems from zero to production at BenQ Materials: architected the AI data-analysis platform the production line uses daily and built the document-AI stack (4 OCR + LLM systems, 1,500+ h/yr saved). Every system is validated against a labelled test set or the system it replaces before go-live.

EXPERIENCE

BenQ Materials Corp. (明基材料) — AI Application R&D Engineer Dec 2024 – Present
Digital Technology Center, Intelligent Application Development Sec. · Taoyuan

AI data-analysis platform ("Enterprise Brain") · FastAPI, Azure OpenAI (company-hosted), Docker — architect & main developer; methods from teammates

- Replaced 6 n8n flows (~550 nodes, logic copied per flow) with **one analysis skill** shared by upload, API and 6 schedules, so a new method is wired once.
- Designed the skill in Claude Code–compatible format (SKILL.md + deny rules) running in a Docker sandbox (non-root, resource/time limits, isolated network), so a future chat agent can call the same methods unchanged; code computes the statistics, the LLM only writes the interpretation.
- Shipped to the polarizer production line, replacing a purchased AutoML package; cut over 6 flows with zero downtime, old vs. new output identical on the same inputs (Excel 9/9 cells, PDFs byte-identical).
- Done: platform, sandbox, OCAP knowledge base (6 defect flows, written by me), feedback labelling (built, rolling out in stages). Prototype only: chat agent, MCP, knowledge graph.

Document-AI systems (OCR + LLM) · Azure Document Intelligence, Azure OpenAI — built solo from zero; 4 systems live

- Shipped quotation, Taiwan/China invoice and health-check OCR; each scored field-by-field in code on a hand-labelled set before go-live.
- Raised the quotation validity field from **51% to 100%** on a 29-document regression set, 30 → 10 s per document; **758+** documents processed in production since.
- Invoices: **31/32** (TW, incl. handwritten) and **31/31** (CN, 3 sites) with every key field correct. Health-check: **97.4%** of fields on 42 documents × 46 fields, 216 processed in production; data encrypted (AES-256-GCM), AD login, ISO 27001-aligned.
- Built an APO (automatic prompt optimisation) tool for the team's OCR projects; in pre-launch validation.

Supplier questionnaire management system (GCP) · React 19, Cloud Functions (Node.js), Entra ID — designed and built solo

- Delivered **52 APIs** and the front end in 4.5 months; live and handed over to the owning team.

ESG carbon & energy platform refactor — took over a 44,000-line legacy platform

- Split a 9,977-line backend file into one module per page (**APIs 104 → 42**, tables 13 → 8); merged 55 copy-pasted chart files into 5 shared components (front end 219 → 51 files).
- Verified each step by API replay and pixel diff of 57 charts, dashboard never down; moved data import from a developer-run script to a web upload owned by the user department.
- Teaching assistant, company ChatGPT Team course (67 learners); Digital Talent L3 (Expert) certification.

China Steel Corporation (中國鋼鐵) — Shift Supervisor (值班股長) Jan 2024 – Nov 2024
Cold-Rolling Mill · Kaohsiung

- Ran shift operations on the cold-rolled coil line at **95% yield**.
- Root-caused defects and downtime with SPC, trend and correlation analysis of process parameters; built the daily anomaly reports (Python / Excel) used in production meetings.

SKILLS

LLM & agents	Azure OpenAI, Gemini API, prompt engineering / APO, skills (Claude Code format), sandboxed execution, function calling, MCP (prototype), golden-set evaluation
Document AI / ML	Azure Document Intelligence, OCR + LLM extraction, PyTorch, YOLOv8
Backend & infra	Python, FastAPI, Node.js, React, JavaScript / TypeScript, PostgreSQL, SQL Server, Docker, GCP, Azure, n8n, Git
Languages	Mandarin (native) · English (professional, TOEIC 790) · German (A2) · military service completed

PROJECTS

Universal OCR Platform · FastAPI, React, Cloud Run · side project 2025 – 2026

- No-code OCR builder: define fields and prompts, deploy as an OCR REST API in one click; document analysis wrapped as MCP-format tools that extract document features and route to the best OCR template.

EDUCATION

National Taiwan University — M.S. Biomechatronics Engineering (AIoT / computer vision) 2021 – 2023

- Thesis (lead developer): greenhouse AIoT pest monitoring and bee pollination — bee tracking **99.5% mAP**, pest counting **95.8%** (YOLOv8, StrongSORT, Mask R-CNN).
- Co-first author (with advisor), *Computers and Electronics in Agriculture* 206:107665 (2023); first author, 10th ISMAB (2022); Taiwan patent TW I842416 B, crop harvester (2024).

National Taipei University of Technology — B.S. Mechanical Engineering · capstone: Autoware / ROS self-driving 2017 – 2021